VeriRAG — Trust-Calibrated Retrieval Evaluation System

---

1. Executive Summary

VeriRAG is a quantitative reliability evaluation framework designed to assess the trustworthiness of AI-generated answers in retrieval-augmented systems.

The system integrates semantic retrieval metrics, alignment scoring, statistical validation, and contradiction detection into a unified Composite Trust Score (CTS) to differentiate grounded responses from hallucinated outputs.

The framework was validated using a large-scale semantic corpus (200K MS MARCO passages) and demonstrated strong statistical separation between grounded and hallucinated responses (ROC-AUC = 1.0).

---

2. Problem Statement

Retrieval-Augmented Generation (RAG) systems improve factual grounding by retrieving relevant context before answer generation. However, existing systems lack:

- Quantitative trust scoring
- Risk-based retrieval diagnostics
- Subtle hallucination detection
- Calibration analysis for confidence reliability

The objective was to design an end-to-end evaluation framework capable of:

- Measuring retrieval quality
- Quantifying semantic alignment
- Detecting contradictions
- Assigning interpretable trust scores

---

3. Dataset & Infrastructure

Corpus: MS MARCO Passage Collection

- 200,000 passages (subset of 8.8M)
- 384-dimensional embeddings
- FAISS cosine similarity index

Tools & Stack:

- Python
- SentenceTransformers (all-MiniLM-L6-v2)
- FAISS
- NumPy / Pandas
- Scikit-learn
- Transformers (NLI model)

---

4. System Architecture

Phase 1 — Retrieval Layer

- Embedding generation (batch encoded)
- FAISS cosine index
- Top-k similarity search

Metrics:

- Retrieval Coverage Score (RCS)
- Retrieval Stability (Std Dev + Top1–TopK Gap)
- Retrieval Risk Classification

Phase 2 — Alignment Layer

- Evidence Alignment Score (EAS)
- Semantic similarity between answer and retrieved context

Phase 3 — Hallucination Modeling

Three-level hallucination detection:

1. **Semantic Gap Risk (RCS − EAS)**
2. **Numeric Claim Verification**
3. **NLI-Based Contradiction Detection**

These were combined into:

Hallucination Probability Score (HPS v3)

Phase 4 — Composite Trust Score (CTS)

CTS integrates:

- Retrieval Coverage
- Evidence Alignment
- Inverse Hallucination Risk
- Stability Score

Weighted aggregation model:

CTS = w1(RCS) + w2(EAS) + w3(1−HPS) + w4(Stability)

---

5. Experimental Evaluation

Grounded vs Hallucinated Testing

Evaluation Type	Avg CTS
Grounded Answers	0.7788
Hard Hallucination	0.3446
Subtle Hallucination	0.6714
NLI-Enhanced Subtle	0.5789

Statistical Validation

- Separation Margin: 0.4341
- ROC-AUC: 1.0
- Classification Accuracy: 1.0
- Threshold Sensitivity Sweep performed
- Calibration via Quantile Binning

No overlap observed between grounded minimum and hallucinated maximum scores.

---

6. Stress Testing

Two hallucination categories were evaluated:

1. Hard hallucinations (fully irrelevant responses)
2. Subtle hallucinations (semantically similar but factually incorrect)

Numeric mismatch penalties and contradiction escalation significantly reduced subtle trust inflation.

---

7. Key Contributions

- Designed multi-metric trust scoring architecture
- Integrated semantic + logical contradiction signals
- Achieved perfect statistical separation on test distribution
- Built scalable evaluation pipeline for large semantic corpora
- Introduced interpretable trust scoring framework

---

8. Limitations

- Synthetic hallucinations were manually generated
- Threshold selection not auto-optimized
- No live LLM integration for generation benchmarking
- External benchmark comparison not performed

---

9. Future Enhancements

- Real LLM answer evaluation
- Automatic threshold learning
- Brier score & advanced calibration metrics
- Cross-dataset benchmarking
- Real-time API deployment

---

10. Conclusion

VeriRAG demonstrates that multi-layered retrieval diagnostics, semantic alignment scoring, numeric verification, and contradiction modeling can be unified into a statistically validated trust scoring framework.

The system successfully differentiates grounded and hallucinated outputs using interpretable metrics and calibration analysis, offering a structured reliability evaluation mechanism for retrieval-augmented systems.